DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY



Report

Database design labwork 5

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Chapter 1

Exercise 1

Which normal form does the employee database satisfy? Why?

The employee database satisfies 3NF (Third Normal Form).

1.1 1NF (First Normal Form)

The requirements of 1NF are:

- Each column contains atomic (single) values.
- There are no repeating groups.
- Each row can be uniquely identified.

In this database, all six tables (`employees`, `departments`, `dept_emp`, `dept_manager`, `salaries`, and `titles`) contain atomic values. Each table has a primary key to uniquely identify its records.

1.2 2NF (Second Normal Form)

The requirements of 2NF are:

- Satisfies 1NF.
- Every non-key attribute depends on the entire primary key (no partial dependency).

Tables with a single-column primary key (`employees`, `departments`) automatically satisfy 2NF.

Tables with composite primary keys (`dept_emp`, `dept_manager`, `salaries`, `titles`) have non-key attributes that depend on the entire composite key.

1.3 3NF (Third Normal Form)

The requirements of 3NF are:

- Satisfies 2NF.
- There are no transitive dependencies (non-key attributes must not depend on other non-key attributes).

Check each table:

Table	**Functional Dependency**
employees	$emp_no \rightarrow birth_date, first_name,$ $last_name, gender, hire_date$
departments	$dept_no \rightarrow dept_name$
dept_emp	$(emp_no, dept_no) \rightarrow from_date, to_date$
dept_manager	$(emp_no, dept_no) \rightarrow from_date, to_date$
salaries	$(emp_no, from_date) \rightarrow salary, to_date$
titles	$(emp_no, title, from_date) \rightarrow to_date$

In each table, non-key attributes depend directly on the primary key. There are no transitive dependencies among non-key attributes.

1.4 3NF of Order form by normalization

Attached in `Information-Systems-2026-2027\lab5\images\lect_5_img_normalize_3NF.jpg`